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SafeFix: Targeted Model Repair via Controlled Image Generation

(/pdf?id=TtpW6JiEiW)

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14 Feb 2026 (modified: 23 Feb 2026) Under review for TMLR Everyone

Revisions (/revisions?id=TtpW6JiEiW) BibTeX

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Abstract: Deep learning models for visual recognition often exhibit systematic errors due to underrepresented semantic subpopulations. While existing debugging frameworks can identify these failure slices, effectively repairing them remains difficult. Current solutions often rely on manually designed prompts to generate synthetic images—an approach that introduces distribution shift and semantic errors, often resulting in new bugs. To address these issues, we introduce SafeFix, a framework for distribution-consistent model repair via controlled generation that employs a diffusion model to generate semantically faithful images that modify only specific failure attributes while preserving the underlying data distribution. To ensure the reliability of the repair data, we implement a verification mechanism using a large vision--language model (LVLM) to enforce semantic consistency and label preservation. By retraining models on the synthetic data, we significantly reduce errors in rare cases and improve overall performance. Our experiments show that SafeFix achieves superior robustness by maintaining high precision in attribute editing without introducing additional bugs.

Submission Type: Regular submission (no more than 12 pages of main content)

Supplementary Material: zip (/attachment?id=TtpW6JiEiW&name=supplementary_material)

Competing Interests: N/A

Human Subjects Reporting: We verify image outputs through a human audit conducted by five AI graduate students on 300 images.

Assigned Action Editor: David Fouhey (/profile?id=~David_Fouhey2)

Submission Number: 7509

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Review of Paper7509 by Reviewer NMkE

Review by Reviewer NMkE 30 May 2026, 02:15 (modified: 30 May 2026, 02:16) Everyone

Revisions (/revisions?id=CRXp0o6kBC)

Summary Of Contributions:
This paper introduces SafeFix, which introduces a pipeline for classifier models against failures related to underrepresented data, which is mentioned as "rare-cases" in the existing dataset. The method consists of multiple steps. Initially, the method identifies classifier failures that can be characterized as high-error and rare cases w.r.t. training data, using attributes proposed by a VLM and VQA-based assessment. Folliwing this step, for the cases that satisfy the failure mode of high-error and low freqnecy, the authors propose a targeted augmentation step using ControlNet with soft HED boundaries as conditioning, which aims to preserve original distribution of the training data, rather than inserting arbitrary samples from the target attribute. Third, the generated images are filtered with LVLMs as a verification step for edit accuracy and preservation of labels. The overall claim of the paper states that since the proposed augmentation prevents degregadion of the overall accuracy of the classifier by preserving the original data distribution except the target attribute (the rar attibute targeted by the augmentation) and does not introduce any new bugs, where systematic classifier errors are defined as bugs by the authors. Authors validate the effectiveness of SafeFix on CelebA (lipstick classification), and 10-class ImageNet subset across ResNet-18, Vit-B/16 and CLIP, compared against six augmentation/generation baselines, primarily via the reduction of the failure rate, with an ablation and a small human audit of the filter.

Strengths

- The method is well-motivated and the design choices are coherent. Conditioning generation on real images via HED structure (rather than free-text prompts) is a reasonable response to the distribution-shift and semantic-drift problems the paper identifies, and adding an LVLM filter directly targets the known unreliability of edits.
- The ablations presented in Table 6 cleanly isolates the CDM and LVLM contributions and shows they are complementary, supporting the core methodological claim rather than asserting it.
- The paper effectively shows the difference between diagnosis and pure augmentation on rare attributes, with analyses presented in Section 5.4
- The human audit grounds the filter's reliability empirically, and reports the failure modes. This also enables a transparent overview on the proposed method.

Weaknesses

- Fix Rate is the dominant metric but is hard to compare across rows, because it normalizes by the initial error rate, which varies a lot between settings. The most striking numbers (e.g., 38.37% on ViT/ImageNet10, base 97.42%) correspond to roughly a one-point absolute accuracy gain, a much smaller real effect than the headline suggests, yet are presented alongside FRs from settings where the same percentage represents many more corrected samples.
- The "no new bugs" claim is defined entirely through the thresholds ρ and ε, whose sensitivity the authors themselves note, so the claim as stated in the main text is stronger than the threshold-dependent evidence supports.
- There seems to be a circular evaluation within the method: The same family of LVLMs proposes attributes, assigns slice membership, filters generated data, and (indirectly) defines the slices on which the "fix" is measured. The human audit validates the filter but not this full loop.
- The empirical baselines are mostly repurposed augmentation methods, while the closest conceptual competitors (Dataset Interfaces, Vendrow et al. 2023; latent-space failure directions, Jain et al. 2023) are discussed but not compared, weakening the "outperforms targeted repair" framing.
- External validity is limited: backbones are trained from random initialization (including CLIP, whose value is its pretraining), and the evaluation rests on one binary CelebA task plus a small 10-class ImageNet subset, which is thin support for claims of cross-architecture, cross-dataset generality. In addition, the performance of the proposed fixing pipleine is questionable for well-trained classifiers, where the impact of SafeFix is not throughly investigated for datasets that are scaled. It makes sense that fixing such models may be more difficult due to the even more complex data distribution but this should be addressed clearly.

Are the claims made in the submission supported by accurate, convincing and clear evidence?: Yes
Explain your answer above:

The claims of the framework are supported by direct controls: the ablation in Table 6 establishes that the CDM and LVLM components contribute complementary gains, and the red-hair vs. yellow-hair comparison on CLIP (Section 5.4) shows that augmenting a diagnosed failure attribute behaves differently from augmenting an undiagnosed-but-rare one, supporting the diagnosis-driven framing. The filter-reliability claim is grounded by a human audit. Across CelebA and ImageNet10, three backbones, and several augmentation scales, SafeFix produces the highest Fix Rate in the large majority of cells in Tables 1 and 2, and Figure 4 shows that the gains concentrate on the targeted attribute slices rather than spreading uniformly, which is consistent with the targeted-repair claim. The evidence would be strengthened by reporting variance over seeds, qualifying the "no new bugs" claim with respect to threshold sensitivity, and broadening evaluation beyond the relatively small CelebA and 10-class ImageNet subset, but the central claims are supported in their current form.

Would at least some individuals in TMLR's audience be interested in knowing the findings of this paper?: Yes
Explain your answer above:
The paper addresses a real and persistent problem: classifiers systematically failing on underrepresented attribute subpopulations, that research across model debugging, robustness, and fairness-oriented augmentation targets. The proposed pipeline is a practical, reusable recipe that can be adopted or compared against without specialized infrastructure, and the individual components are easy to swap as better diffusion models and LVLMs become available. Furthermore, SafeFix offers a flexible recipe that can be adopted in future and thus may researchers can furher develop more comprehensive validation schemes on top of this study.

Requested Changes:

- The authors should consider softening "no new bugs" claim from the perspective of hyperparameter sensitivity.
- For the evaluation, the authors should utilize non-overlapping VLMs in the framework and evaluation. This would show that the measured fix is not partly an artifact of the same model used throughout the pipeline.
- An empirical comparison against at least one closely related targeted-repair-via-generation method (Vendrow et al. 2023, Dataset Interfaces; or Jain et al. 2023, latent-space failure directions) can be added. These are discussed in related work but not run, and they are the natural competitors for the central claim that SafeFix outperforms targeted-repair approaches.
- For the validity of the method against large-scale datasets and pretrained classifiers, the authors should either utilize their method on classifiers that are trained on larger scale data (such as the complete ImageNet dataset) or perform their training accordingly (on larger datasets). This would enable understanding if the method is limited with large datasets, or can still resolve bugs in larger scale models. The current setup may understate the difficulty of repairing models whose failure modes arise from more complex data distributions, and this limitation should be addressed directly rather than left implicit.

Broader Impact Concerns:
The potential biases on demographics has been mentioned also by the authors in the limitations section. On top of this, there seems to be no addiitional issues worth mentioning.

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Review of Paper7509 by Reviewer KtnS

Review by Reviewer KtnS 30 Apr 2026, 16:02 (modified: 30 May 2026, 02:16) Everyone
Revisions (/revisions?id=voNzf87ZtP)

Summary Of Contributions:
The paper proposes a pipeline to improve classifier performance on rare semantic attribute compositions by generating additional samples with the rare compositions and training the model on these samples.

Are the claims made in the submission supported by accurate, convincing and clear evidence?: Yes
Explain your answer above:
The claim is supported, but it is not a contribution. I am also not sure about its novelty.

To my understanding, the claim in the paper is the design of the pipeline to improve the performance of classifiers on rare semantic subpopulations. But the pipeline is almost directly from prior works, with some modifications to strengthen individual components. For example, compared to HiBug (Chen et al., 2023), this work converts the generation process to a more controllable one w.r.t. the attributes using another prior work, ControlNet. While the use of ControlNet is still fine as a reusable module that is not claimed as part of the contributions, the overall pipeline feels like a mix of existing works. Another example of a reused idea is filtering using LVLMs. This is also a well-known idea, particularly in dataset generation.

Would at least some individuals in TMLR's audience be interested in knowing the findings of this paper?: Yes
Explain your answer above:
Yes, this work is definitely of interest to ML audience.

Requested Changes:

1. I am still okay with this work if the pipeline can be sufficiently justified as a contribution. If I have misunderstood the prior works and their connection to this manuscript, please clarify it in the paper, preferably using a table with columns showing ways in which the manuscript differs from the prior works. In the table, please mention the high-level ways in which the proposed pipeline differs from those in the prior works (instead of "they use text-to-image models, while we use image-conditioned models").
2. According to the paper, the metric Fix Rate "measures the fraction of previously misclassified samples that are corrected." However, Eq. (6) does not compute that. It looks only at the aggregate accuracy, not per-sample accuracy.
3. This is a question, not a requested change. You mentioned that works like AdaVision and DCD retrieve samples from another dataset. If you have a sufficiently rich dataset, how do such methods compare to the proposed generation-based approach?

Broader Impact Concerns:
N/A

Add: **Official Comment**

Inquiry on Review Status

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Official Comment
by Authors (Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2), RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2))
20 Apr 2026, 02:14 Editors In Chief, Action Editors, Authors

Comment:
Dear Action Editor,

I hope this message finds you well.

I am writing to kindly inquire about the current status of our submission (Paper ID 7509) on OpenReview. It has been about one month since the first review was posted, and we are still awaiting the remaining reviews.

We understand that the review process can take time, and we truly appreciate the effort of the reviewers. We would be very grateful if you could kindly follow up with the remaining reviewers, or share any updates on the timeline and when we might expect the remaining reviews.

Thank you very much for your time and support.

Add: **Official Comment**

Official Comment by Action Editor 1UUt

Official Comment by Action Editor 1UUt (David Fouhey (/profile?id=~David_Fouhey2))
01 May 2026, 14:25 Editors In Chief, Action Editors, Authors

Comment:
Dear Authors,

Thank you for your patience. I have a second review in, and I'm waiting on the last review. Unfortunately a few of the initial assignments ended up not working out for various reasons.

Best wishes,

David

Add: **Official Comment**

Review of Paper7509 by Reviewer 1vpz

Review by Reviewer 1vpz 18 Mar 2026, 18:37 (modified: 30 May 2026, 02:16) Everyone
Revisions (/revisions?id=hGydx9yLfW)

Summary Of Contributions:
SafeFix is a pipeline for fixing vision classifiers that fail on rare attribute subgroups. It has four stages:

1. diagnose rare-case failure slices using HiBug-style attribute-based analysis,
2. generate synthetic images for those slices using ControlNet conditioned on HED edges from real images,
3. filter generated images with an LVLM (Qwen2.5-VL) to verify attribute correctness,
4. augment the training set and retrain. Experiments on CelebA (binary lipstick classification) and a 10-class ImageNet subset show Fix Rate improvements over six baselines across ResNet, ViT, and CLIP.

Once the failing subgroup is identified (via HiBug), the remaining task is generating more data for a rare slice and retraining, which is similar to long-tail learning with synthetic augmentation. The paper's contribution over HiBug is using a better generator (ControlNet instead of unconditioned text-to-image) and adding a quality filter (LVLM).

Strengths:

- The LVLM filtering step addresses a real problem: diffusion models frequently get target attributes wrong.
- Section 5.4 (red-hair vs. yellow-hair augmentation on CLIP) shows targeting the correctly diagnosed bug matters.
- Clear writing, clean algorithm box, good visual comparison.

Weaknesses:

- The contribution reduces to two incremental upgrades over HiBug: swap the generator, add a filter.
- The paper does not engage with the long-tail learning field. And the scalability is not well presented.

Are the claims made in the submission supported by accurate, convincing and clear evidence?: No
Explain your answer above:

1. The contribution is not clearly separated from prior work The diagnosis stage (Stage 1) is HiBug's algorithm. The retraining stage (Stage 4) is standard. The actual difference is in Stages 2–3: a stronger generation backbone (ControlNet instead of plain text-to-image) and the addition of LVLM filtering. The delta over HiBug is primarily "use stronger off-the-shelf tools", which is an incremental upgrade enabled by today's foundation models.
2. The paper does not engage with the long-tail learning field. And the scalability is not well presented. Once HiBug identifies a rare failing slice, the remaining challenge reduces to: how best to augment an underrepresented subgroup? This is fundamentally a long-tail learning problem, yet the paper makes no connection to this well-established field. In particular, the paper does not adequately analyze whether injecting synthetic data for rare slices shifts the effective training distribution and degrades performance on common classes. By the standards of the long-tail learning literature (which is concerned precisely with improving subgroup performance without compromising majority-class accuracy, and is thus essentially distinct from few-shot learning), the scale of evaluation presented here is insufficient to convincingly demonstrate distributional safety. In particular, both datasets are not large enough to support the paper's generality and scalability: CelebA is reduced to a single binary task with a small attribute vocabulary, and ImageNet10 contains only 10 hand-picked classes. This is

particularly concerning given the long-tail framing, where the challenge scales with the number of classes and the complexity of attribute interactions. Demonstrating the pipeline on a setting with a larger tail and richer attribute space is necessary to make the claims convincing.

Would at least some individuals in TMLR's audience be interested in knowing the findings of this paper?: Yes
Explain your answer above:
The problem, fixing classifiers on underrepresented subgroups, is important for fairness and deployment reliability. Using LVLMs as quality gates for synthetic augmentation is a timely idea. The diagnose-generate-filter-retrain pipeline is feasible and could be adopted by practitioners. Researchers in subgroup robustness, data-centric AI, long-tail learning, and fairness-aware training would find the framing relevant.

Requested Changes:

1. Clarify the paper's contribution.
2. Engage with the long-tail learning literature (or discussing why not), since the core task after diagnosis is augmenting underrepresented subgroups without degrading majority-class performance, which is the central problem of that field.

Broader Impact Concerns:
The pipeline generates synthetic faces with targeted demographic attributes (skin tone, hair color, gender). But if the LVLM rejects brown-skin images at a higher rate than white-skin images (plausible given known biases in vision-language models), the "fairness repair" pipeline may itself be biased. Similarly for the intrinsic bias in the generation model.

Add: **Official Comment**

Review Approval of Paper7509 by Action Editor 1UUt

Review Approval by Action Editor 1UUt (David Fouhey (/profile?id=~David_Fouhey2))
23 Feb 2026, 18:11 (modified: 23 Feb 2026, 18:11) Editors In Chief, Action Editors, Authors
Revisions (/revisions?id=QGK1KqS1Ad)
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